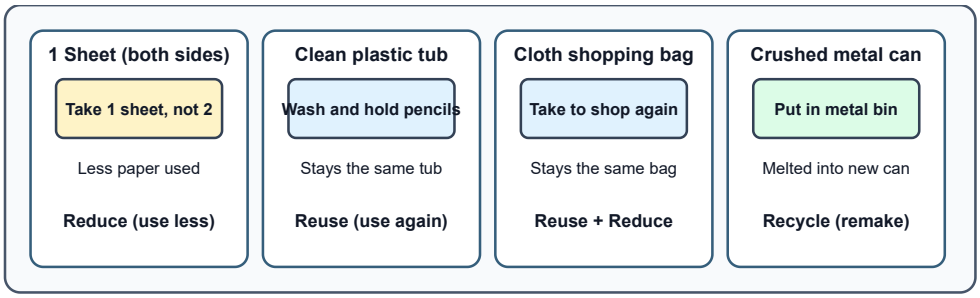
Reference sheets

Reference A · Classroom waste examples

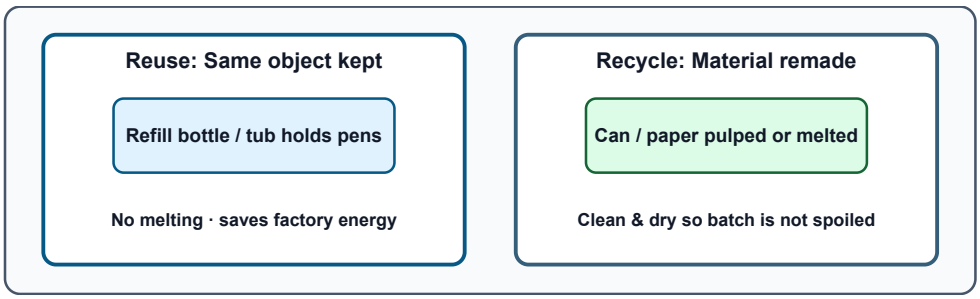


Four classroom waste items compared across action, object clue, and waste route.

Reference only. Do not write on this sheet. Use both clue columns to choose reduce, reuse, or recycle.

Classroom item	What we do with it	What happens to the object	Waste route	Why it helps
One sheet of paper (both sides)	Write on side 1 and side 2	We take 1 sheet instead of 2 at the start	Reduce	Saves new paper before waste is made
Clean plastic tub	Wash it and store pencils on the desk	Stays the same plastic tub	Reuse	No new pot needed and no factory melting
Cloth shopping bag	Carry shopping in it many times	Stays the same cloth bag	Reuse (and Reduce)	Replaces single-use plastic bags
Crushed metal drink can	Put clean and dry into the metal bin	Melted at a factory into new metal	Recycle	Saves digging up new metal ore

Reference B · How to tell reuse and recycle apart



Reuse keeps the same object without melting; recycle remakes clean, dry material at a factory.

Check question	Reuse	Recycle
Does the item stay the same object?	Yes — the tub stays a tub; the bottle stays a bottle	No — the can or paper is broken down and remade
Does it need factory melting or pulping?	No — wash or refill and use again straight away	Yes — uses factory heat, water, and energy
What preparation is needed?	Keep clean and safe to use again	Empty, rinse, and dry so food does not spoil the batch

Made by Matt

Science · BUILD · Autumn 2 · Week 1 · SCI_BUILD_A2_W01_P1

Tier pack · Supported

Use Reference A and Reference B. Work in your book or on this sheet. Point, speak, sign, draw, type, write, or tell an adult. You may pass.

Arrival · 4 questions

1. Look at the picture. Which tray shows a clean tub holding pencils: Tray A or Tray B?

Help: Look at Tray B on the right side of the picture.

2. Think back: where do animals get their food?

Help: Choose: from what they eat, or from sunlight alone.

3. Think back: what pulls on your bones to help you move?

Help: Point to your arm. Muscles pull on bones.

4. Look at Tray A. The paper has a blank back. Can we write on the blank back?

Help: Choose Yes or No. A blank back has space for writing.

You do tasks

1. Look at Reference A. Match the paper with a blank back to Reduce (use less) or Recycle. Staff I ☐ S ☐ C ☐

Help: Writing on the back means we take less new paper.

2. Pick one clean item from Reference A that we can use again (reuse). Say what it can hold. Staff I ☐ S ☐ C ☐

Help: Look at the clean tub or the cloth bag in Reference A.

3. Complete this sentence: I can use the ___ again because it is ___ and safe. Staff I ☐ S ☐ C ☐

Help: Choose tub or cloth bag, and use the word clean.

Exit ticket

1. I write on the blank back of my sheet. Is that Reduce (use less) or Recycle?

2. Can we use a clean, safe tub again to hold pencils?

3. What is one thing we check before putting a clean can in a recycling bin?

Made by Matt

Teacher answers and topic guidance

Accept any scientifically equivalent wording in spoken, signed, pointed, drawn, or written form.

Starter answer

Option A (Reuse). Washing a plastic tub and keeping pencils in it uses the same tub again without melting or remaking the plastic.

Arrival · Supported

1. Tray B shows a clean tub holding pencils.
2. Animals get food from what they eat.
3. Muscles pull on bones to help us move.
4. Yes. We can write on the blank back to use less paper.

Arrival · Independent

1. Tray A shows a paper sheet with a blank back.
2. Animals get food from what they eat, while green plants make their own food using light.
3. The skeleton holds the body up and muscles pull on bones so we can move.
4. Using the clean tub again holds pencils without needing to buy or make a new tub.

Arrival · Aspire

1. Tray A uses the blank back so we take no new paper, and Tray B reuses a clean tub so we need no new pot.
2. Animals cannot make their own food, so they must eat to get energy and materials for their bodies.
3. A muscle can contract and pull a bone across a joint, so another muscle pulls it back.
4. We must check that the tub is clean and has no sharp or broken edges.

You do · Supported mark scheme

1. The paper with a blank back matches Reduce (use less). *(Up to 2 marks)*
2. We can reuse the clean tub to hold pencils, or reuse the cloth bag to carry books. *(Up to 2 marks)*
3. I can use the tub again because it is clean and safe. *(Up to 2 marks)*

You do · Independent mark scheme

1. Reduce: sheet with blank back. Reuse: clean tub and cloth bag. Recycle: empty metal can (after checking local bin rules). *(Up to 3 marks)*
2. Using the cloth bag again is reuse. It saves new material because we do not need to make a fresh bag. *(Up to 3 marks)*
3. First check that the item is clean and has no sharp edges. Second check the local bin label to see if that material is accepted. *(Up to 2 marks)*

You do · Choice mark scheme

1. 1. Reduce (use less). 2. Reuse (use again). *(Up to 2 marks)*
2. 3. Metal. 4. Local bin label. *(Up to 2 marks)*
3. Use a clean cloth bag again because it is clean and needs no new material. *(Up to 2 marks)*

You do · Aspire challenge mark scheme

1. Reusing keeps the same clean tub and uses it again without changing it. Recycling breaks the old plastic down at a factory to make a new item. *(Up to 2 marks)*
2. Using less and reusing a safe item stop waste right away without needing lorries and factory energy to melt and remake the material. *(Up to 2 marks)*

Exit ticket · Supported

1. That is Reduce (use less).
2. Yes. That is reuse.
3. We check the local bin label.

Exit ticket · Independent

1. Reduce: write on the blank back of a sheet. Reuse: fill a clean tub with pencils or use a clean cloth bag again.
2. Material means what a thing is made of, such as metal in a can or paper in a sheet.
3. Different places accept different materials, and the wrong item can spoil the recycling load.

Exit ticket · Aspire

1. Reuse uses the same clean item again as it is, while recycle processes old material into a new product.
2. Refilling reduces the number of new bottles made and avoids the energy needed to collect and reprocess five bottles.
3. An unchecked waste item could be dirty or have sharp edges that could hurt someone.

Topic guidance and common misconceptions

- **Reuse vs recycle:** Pupils often call any positive environmental action 'recycling'. Emphasise the object test: if the bottle or tub stays the same object, it is **reuse**; if a factory melts or pulps the material to make a new thing, it is **recycle**.
- **Why reduce and reuse come first:** Recycling still uses collection trucks, heat, water, and electricity. Reducing and reusing save both raw materials and factory processing energy.
- **Clean and dry sorting:** Food or liquid left inside containers can spoil a whole load of paper or card in a recycling bin.

Made by Matt